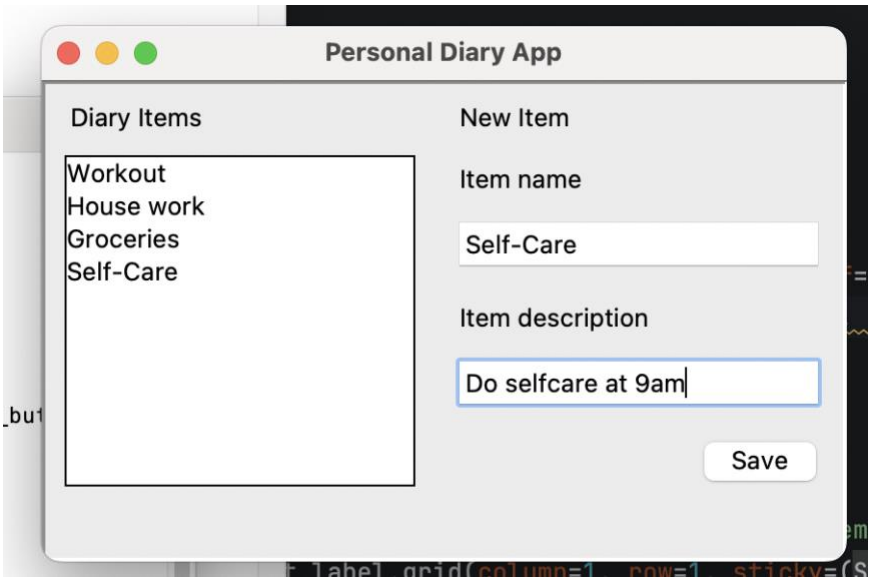
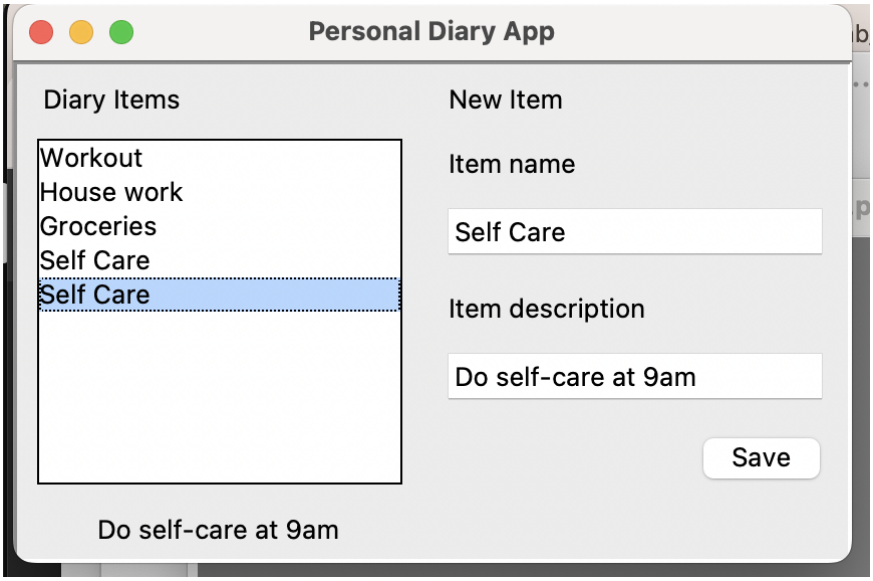
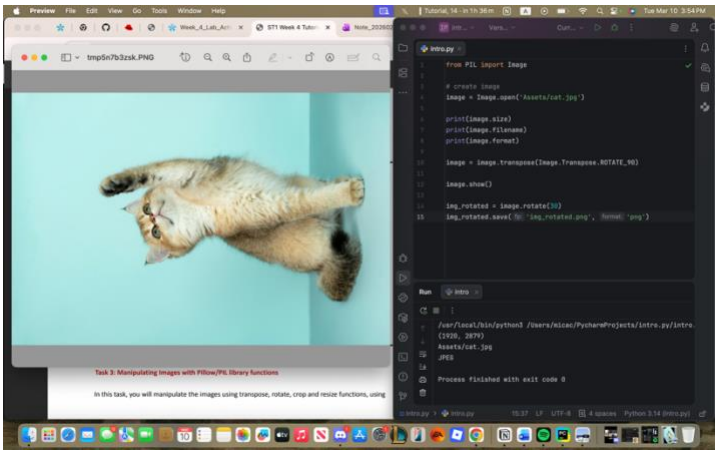
<u>Task 1:</u> Introduction to Tkinter

Try running tkinter_practice.py and personal_diary_app.py from week_4_Lab_Activites folder and understand how to build a Graphical User Interface using different Tkinter widgets. Show the screenshots for a successful test run of personal_diary_app.py below:



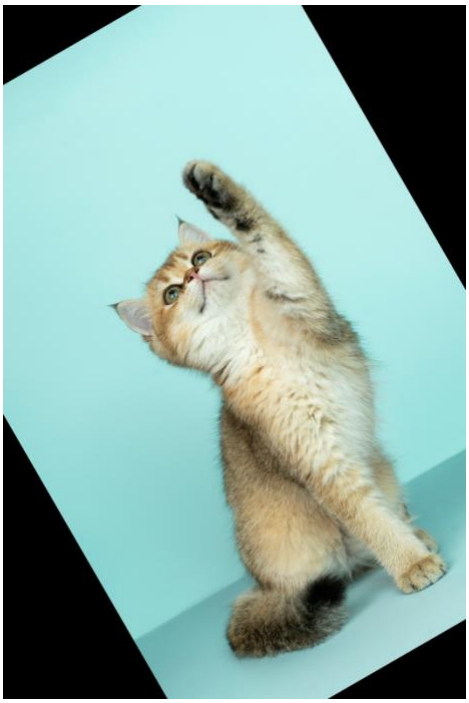


<u>Task 2:</u> Introduction to Pillow library/package

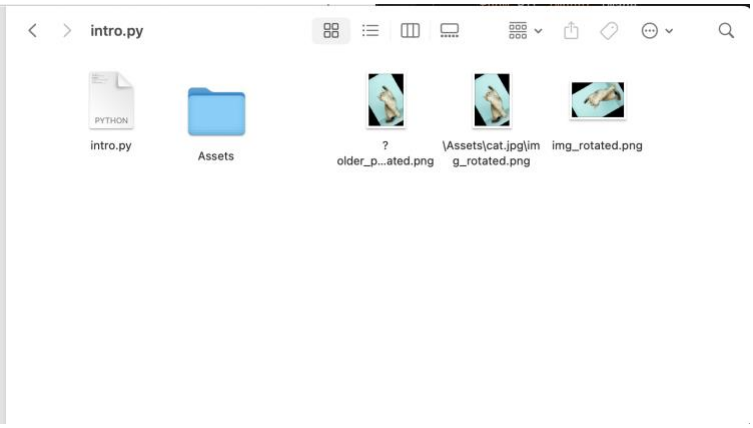


1) <u>Show what does this code display?</u>

This code displayed some of the basics of PIL. Here, you can find the details of the image, (e.g. size, filename, and format of the image) as well as being able to open the image rotated, and saving the image to file while being rotated to a certain degree.



- This is the image used, rotated at a 30-degree angle and saved into file.



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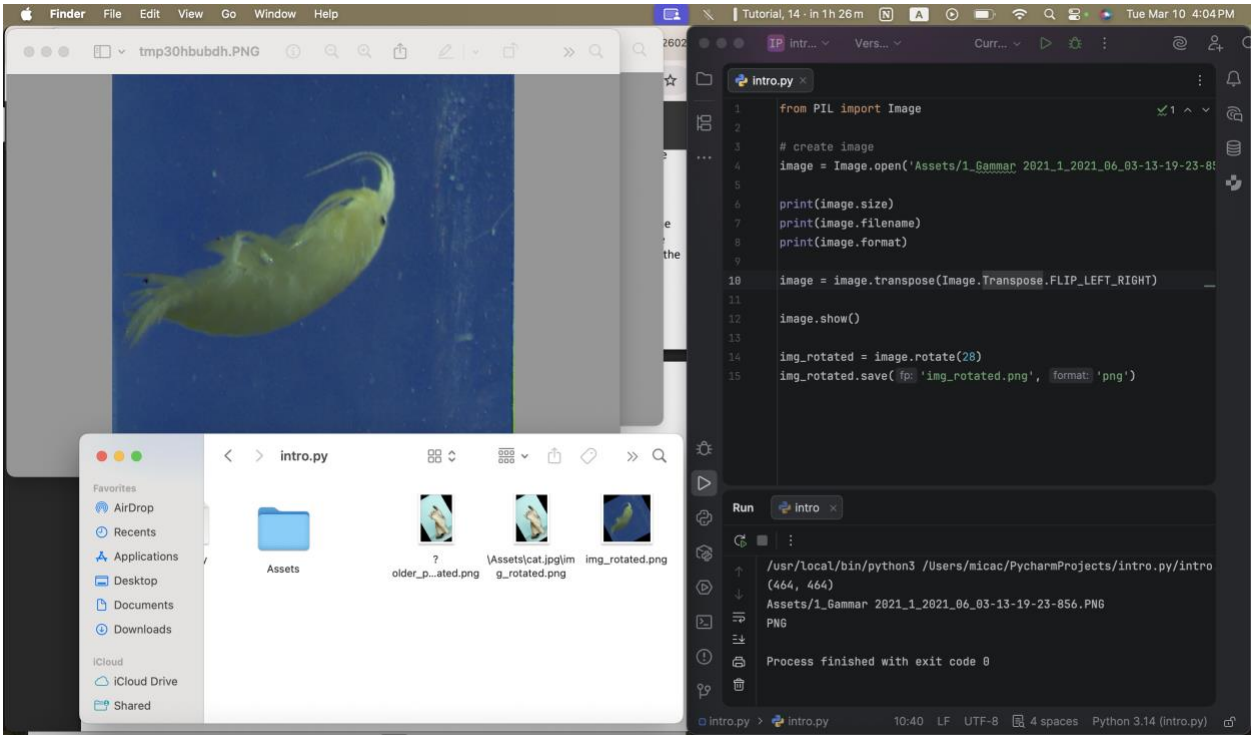

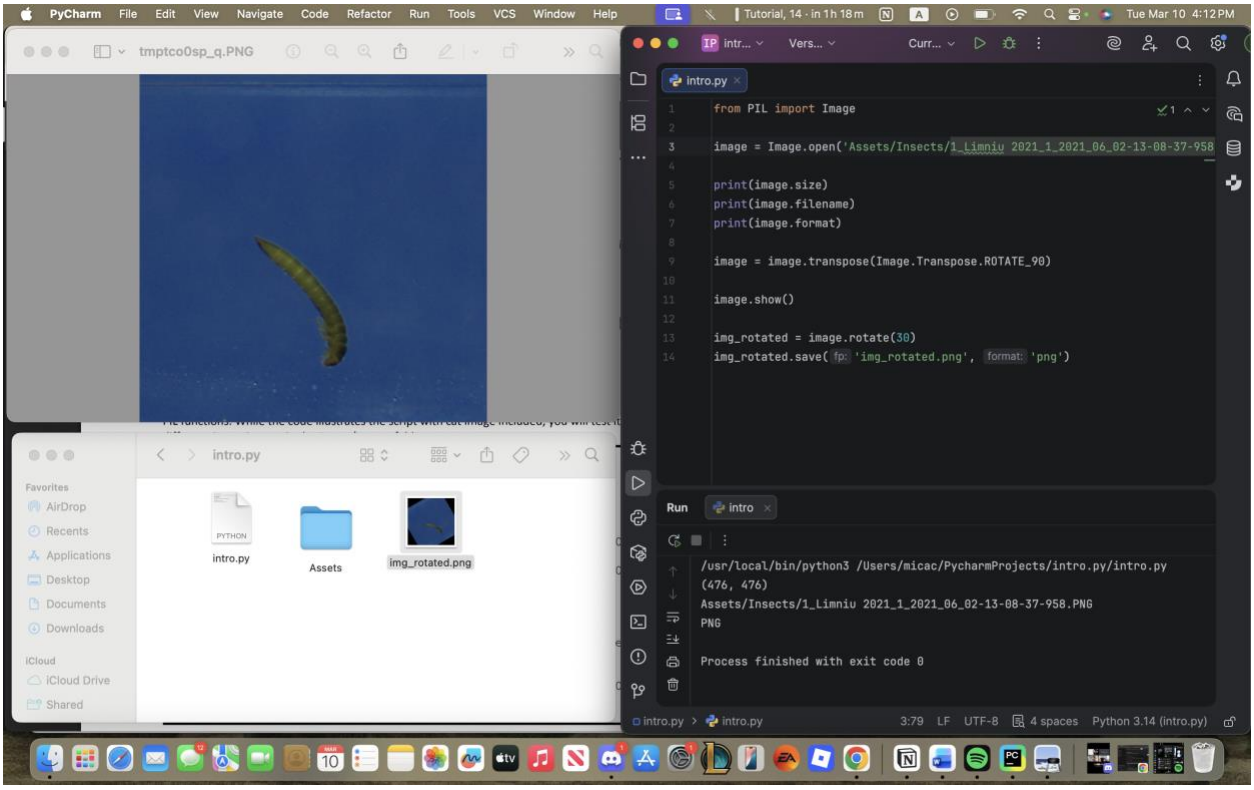

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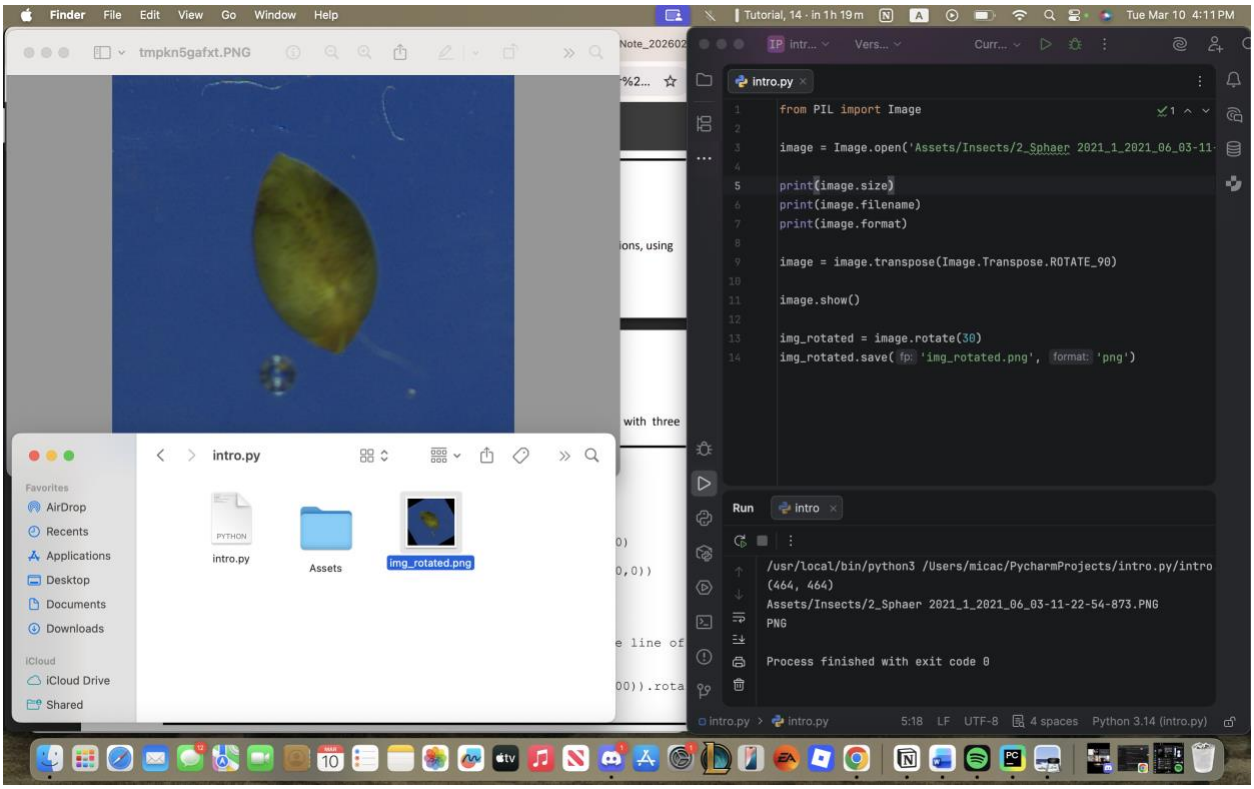
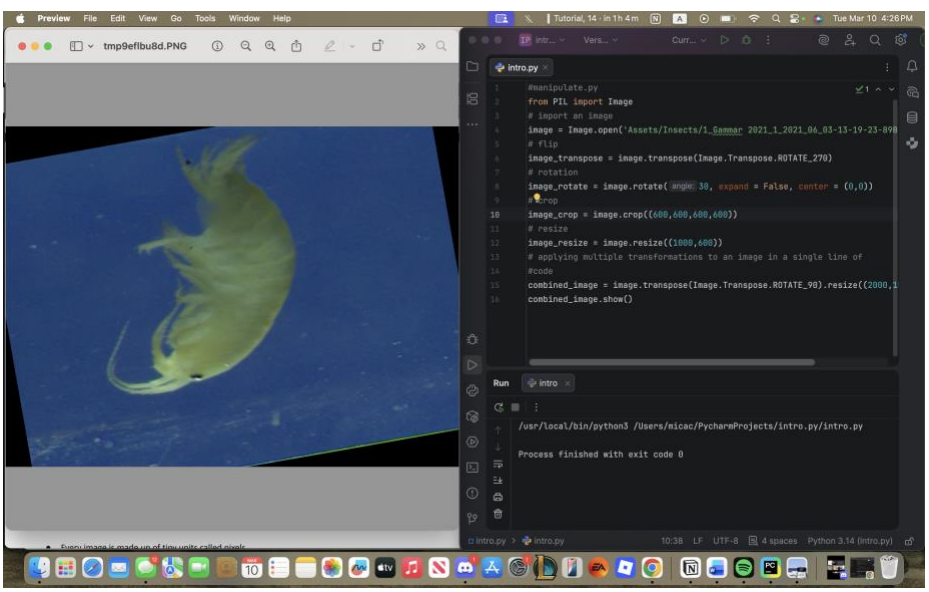


This is my script with a couple of the insects from the assets folder.

<u>Task 3:</u> Manipulating Images with Pillow/PIL library functions



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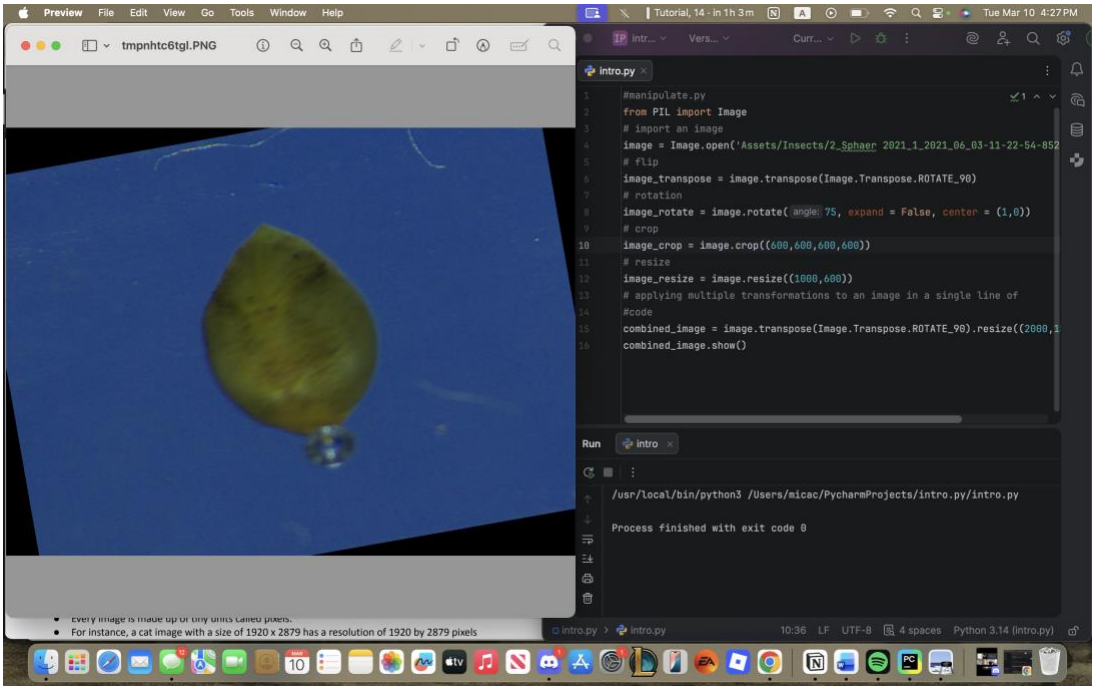

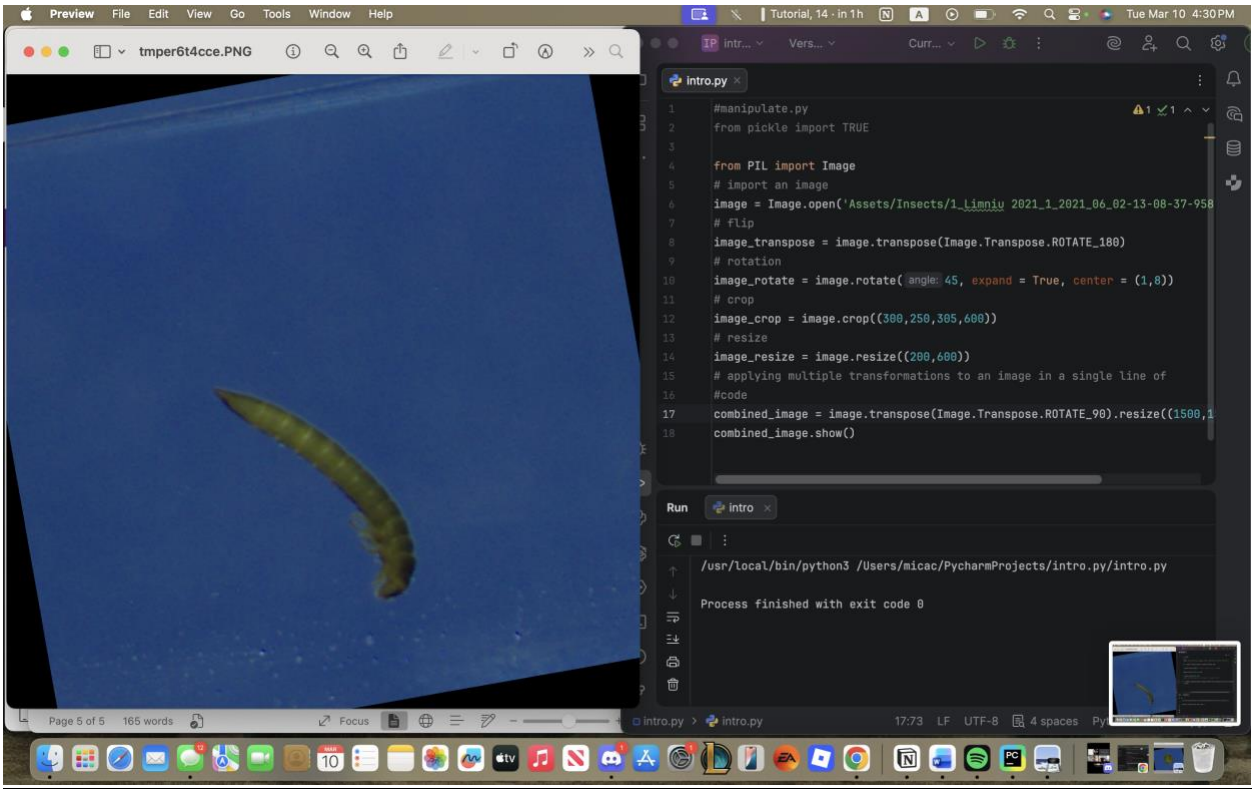

<u>Task 4:</u> Understanding Pixels and Colors with Pillow/PIL library functions

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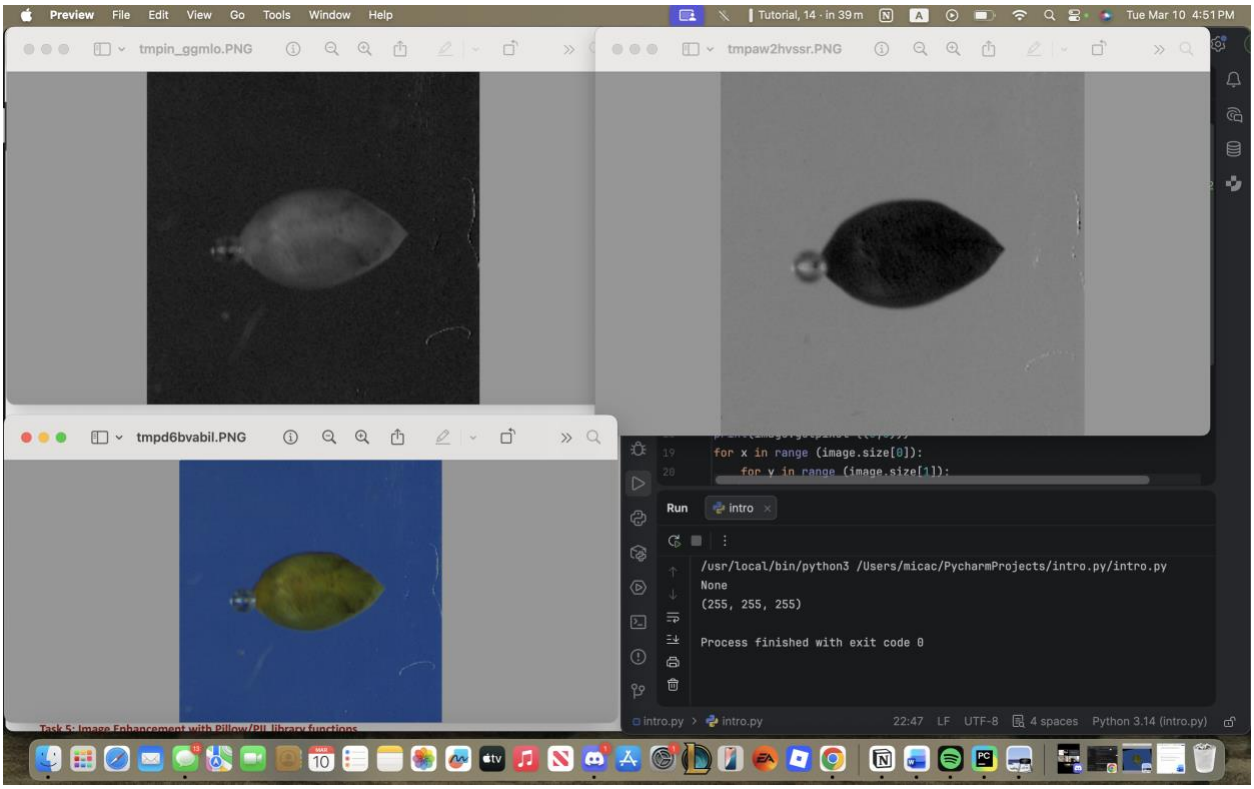

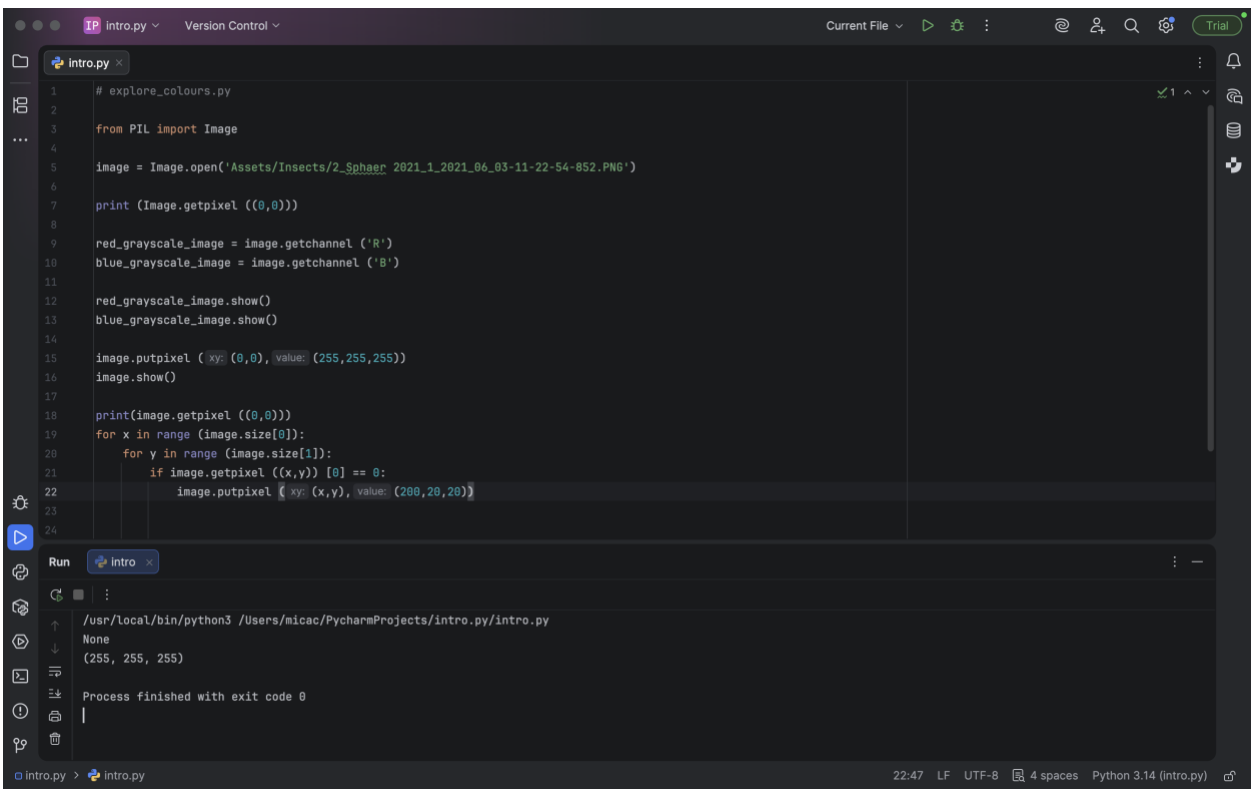

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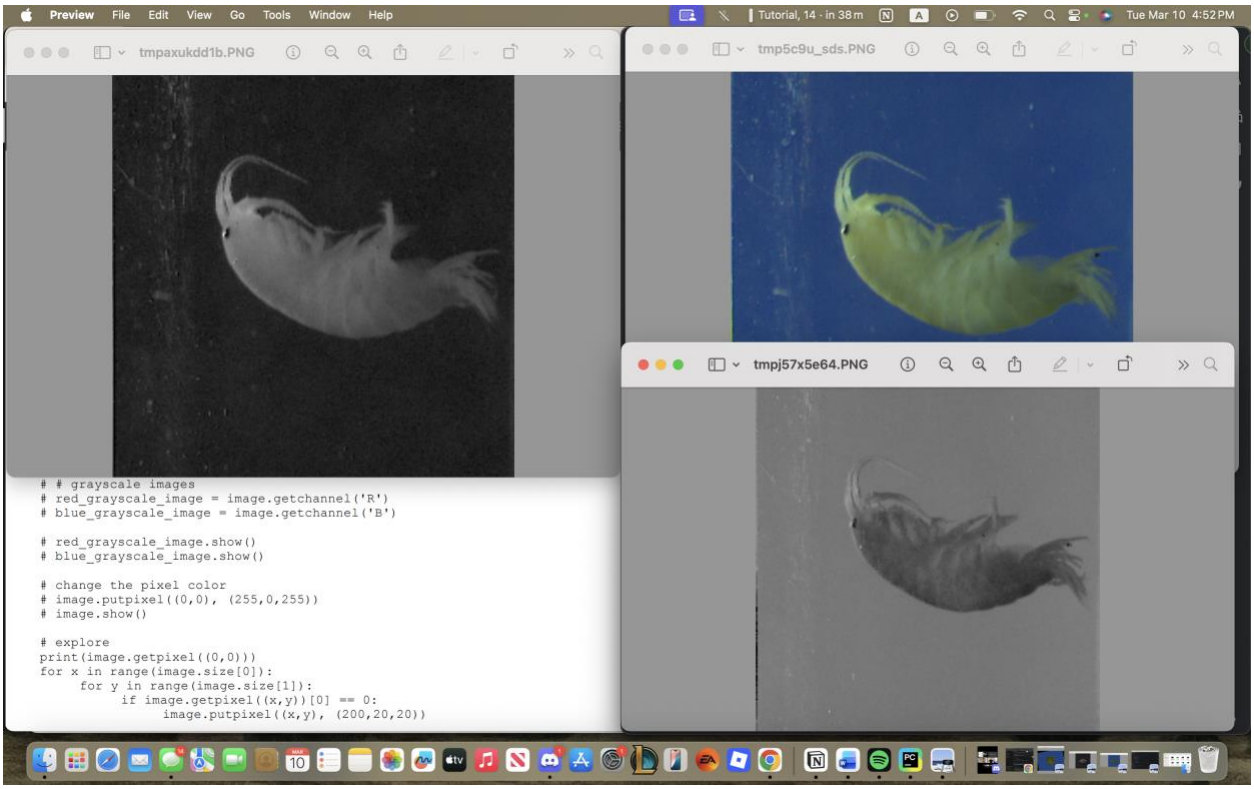

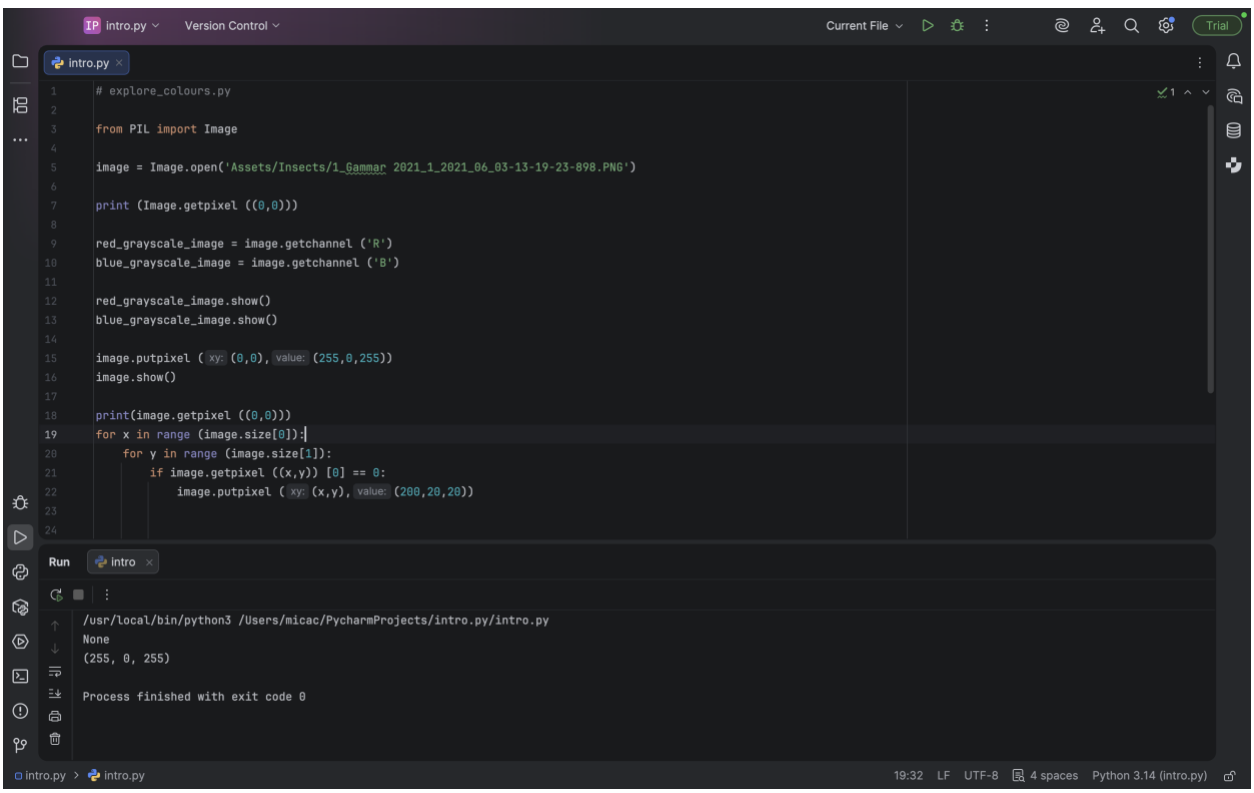

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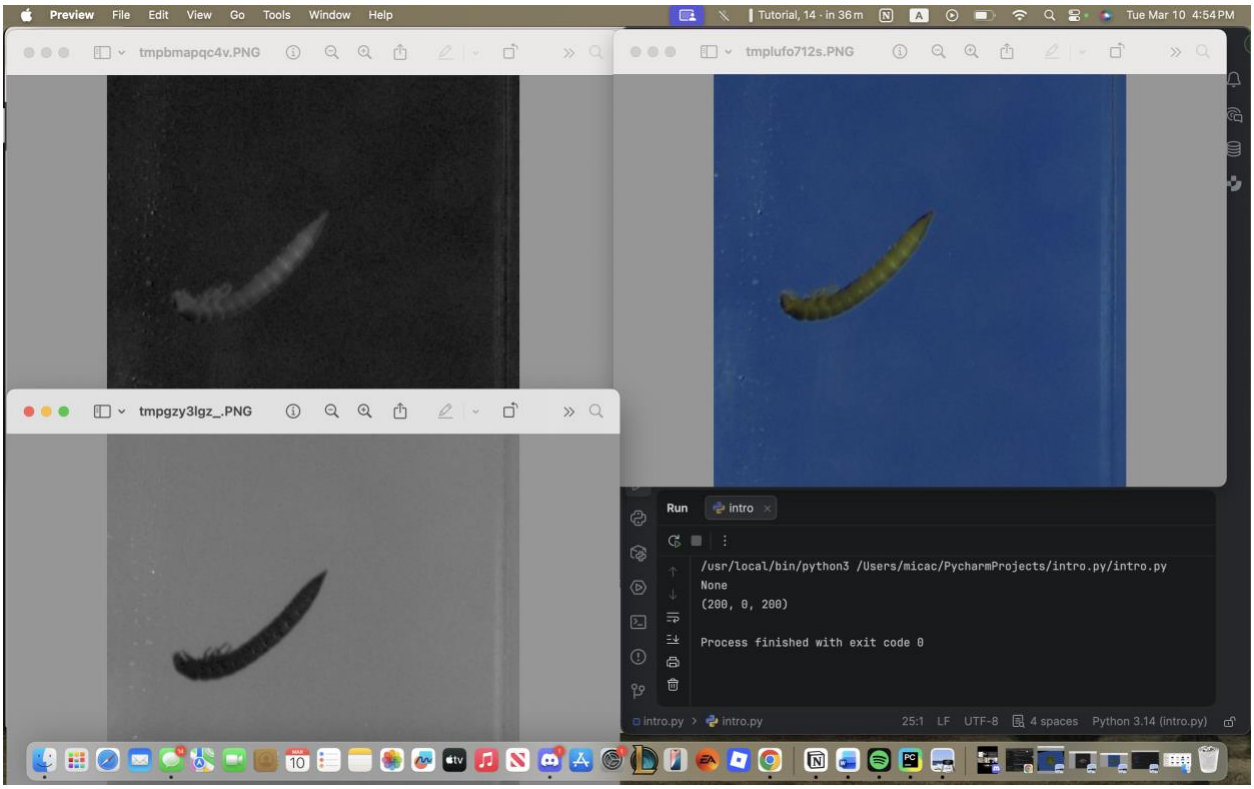

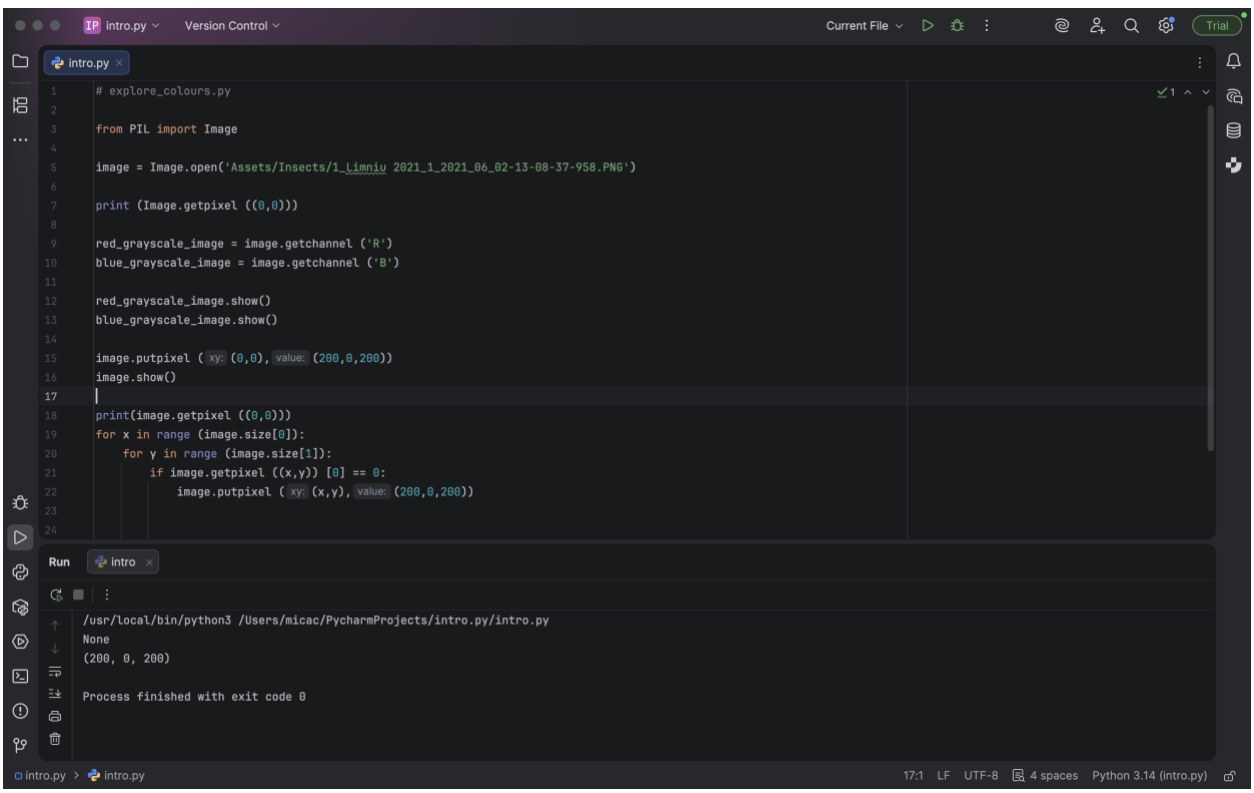

<u>Task 5:</u> Image Enhancement with Pillow/PIL library functions

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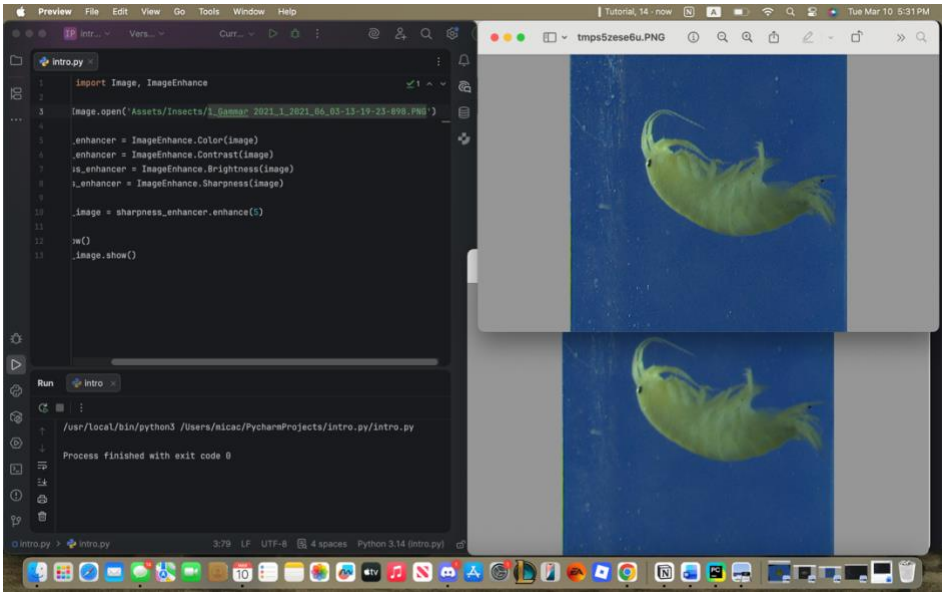
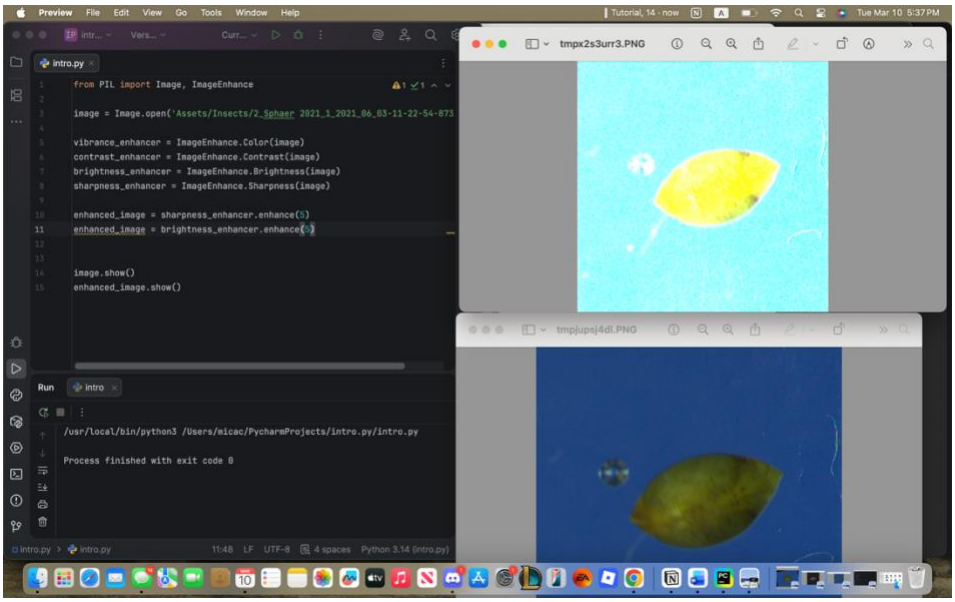
Top photo would be the enhanced photo.



Top photo would be the enhanced photo (sharpness 5)



Top photo would be the enhanced photo

(Brightness 5, Sharpness 5)

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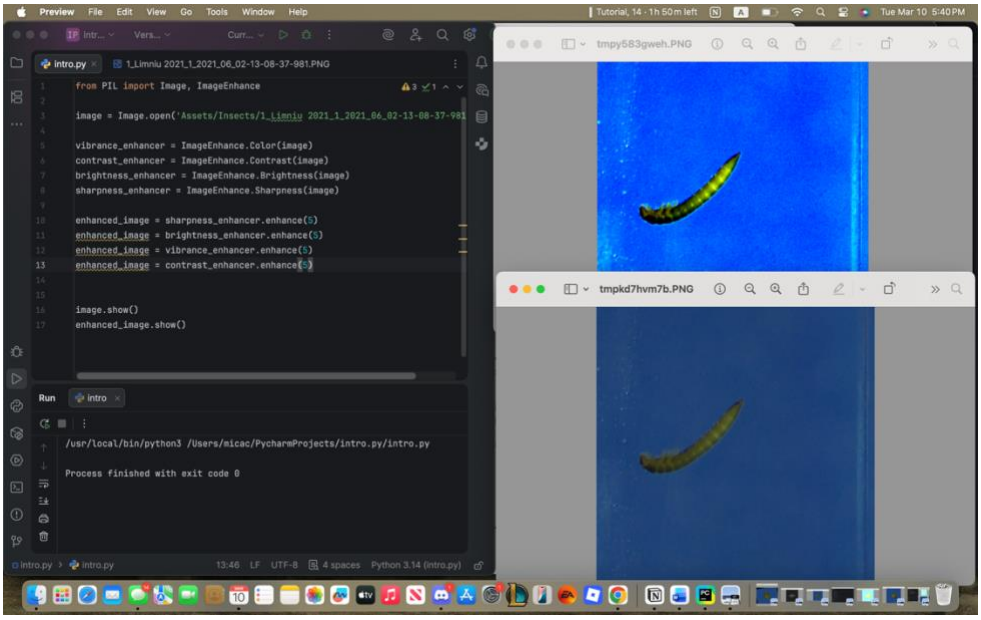
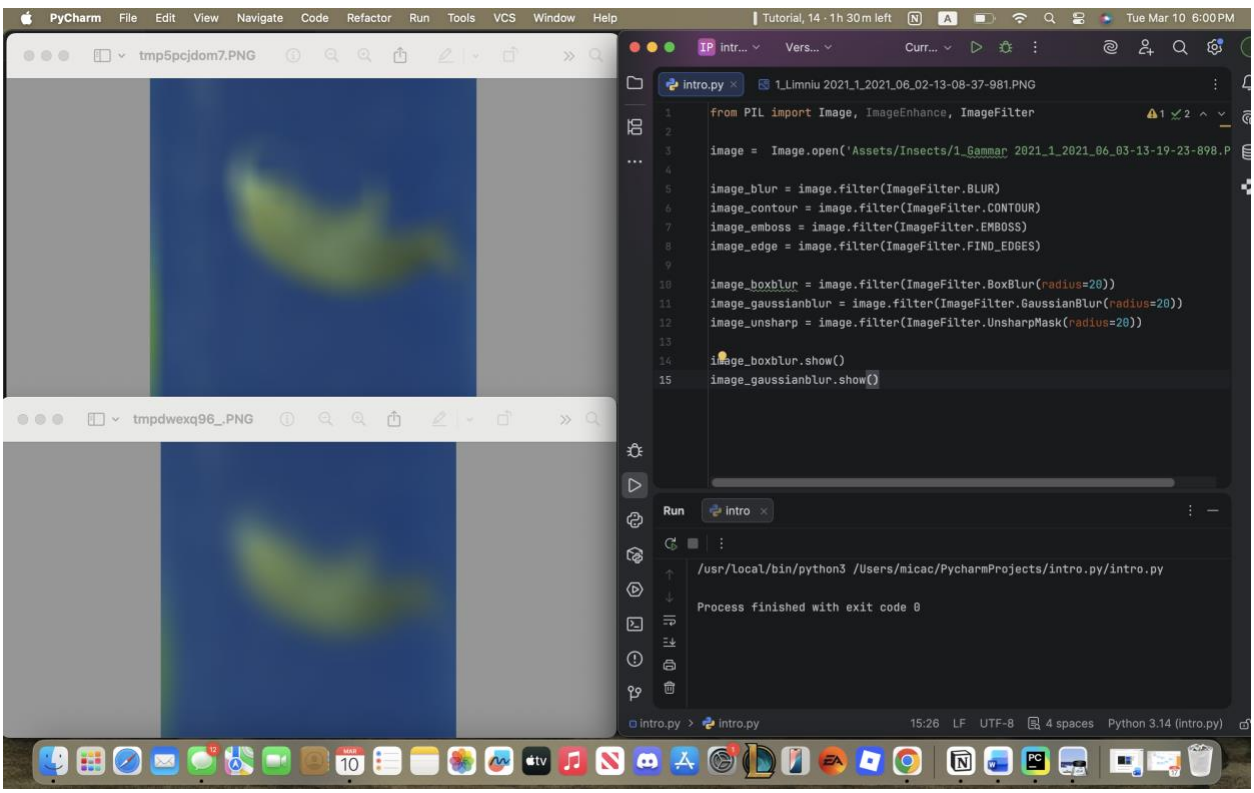


Top photo would be the enhanced photo

(Brightness 5, Contrast 5, Sharpness 5, Vibrance 5)

<u>Task 6:</u> Image filtering and Edge detection with Pillow/PIL library functions



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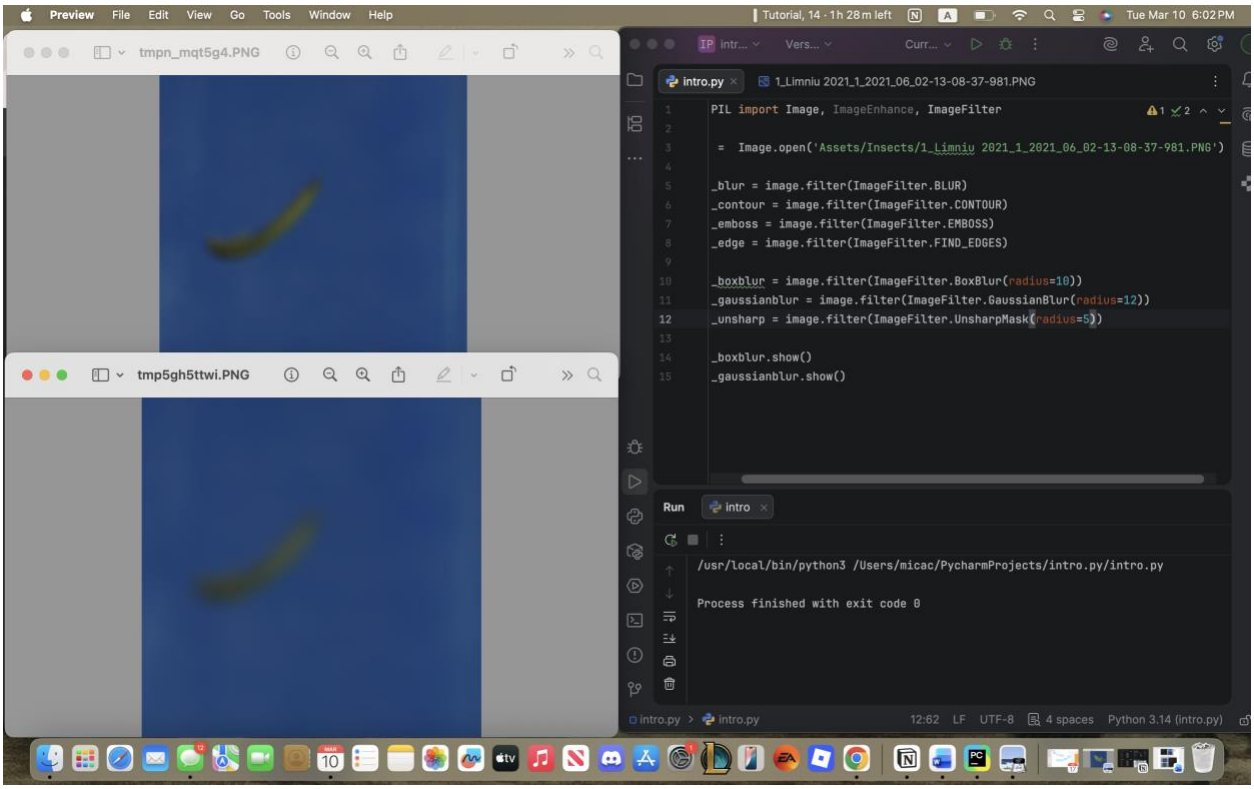

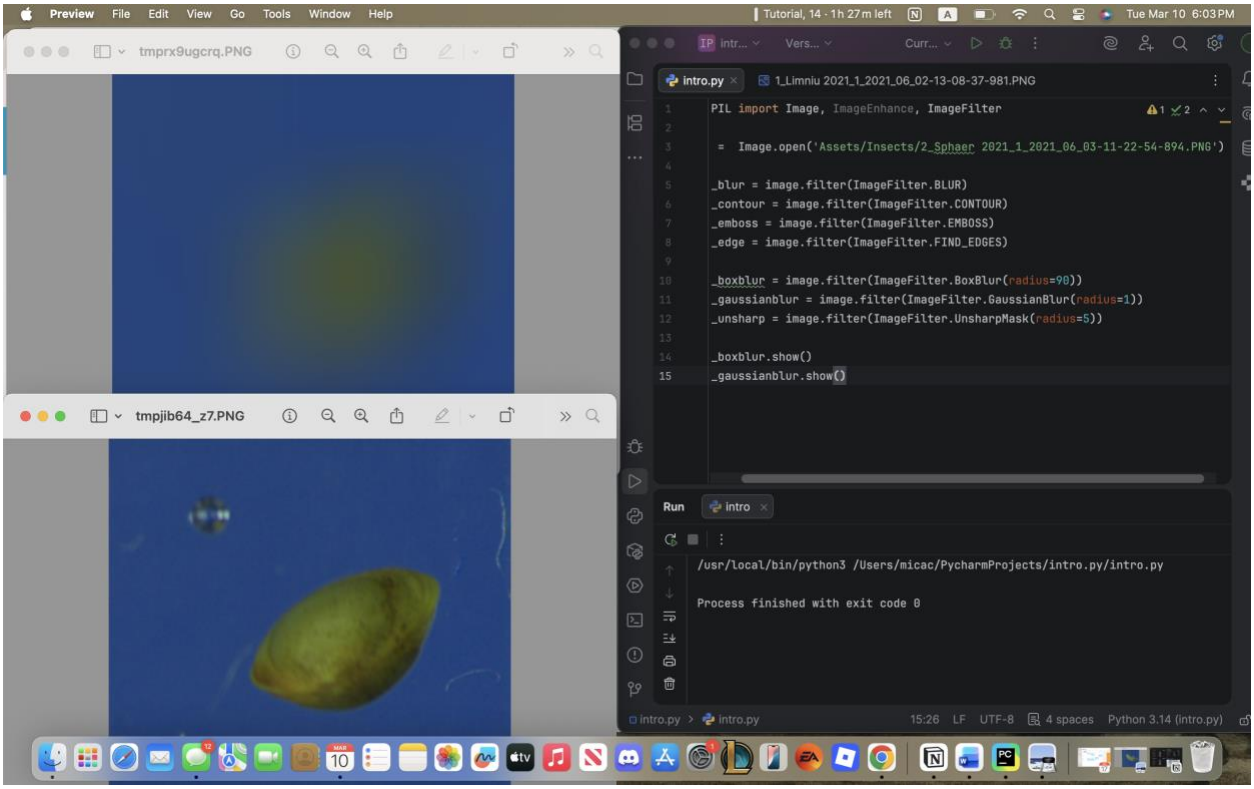

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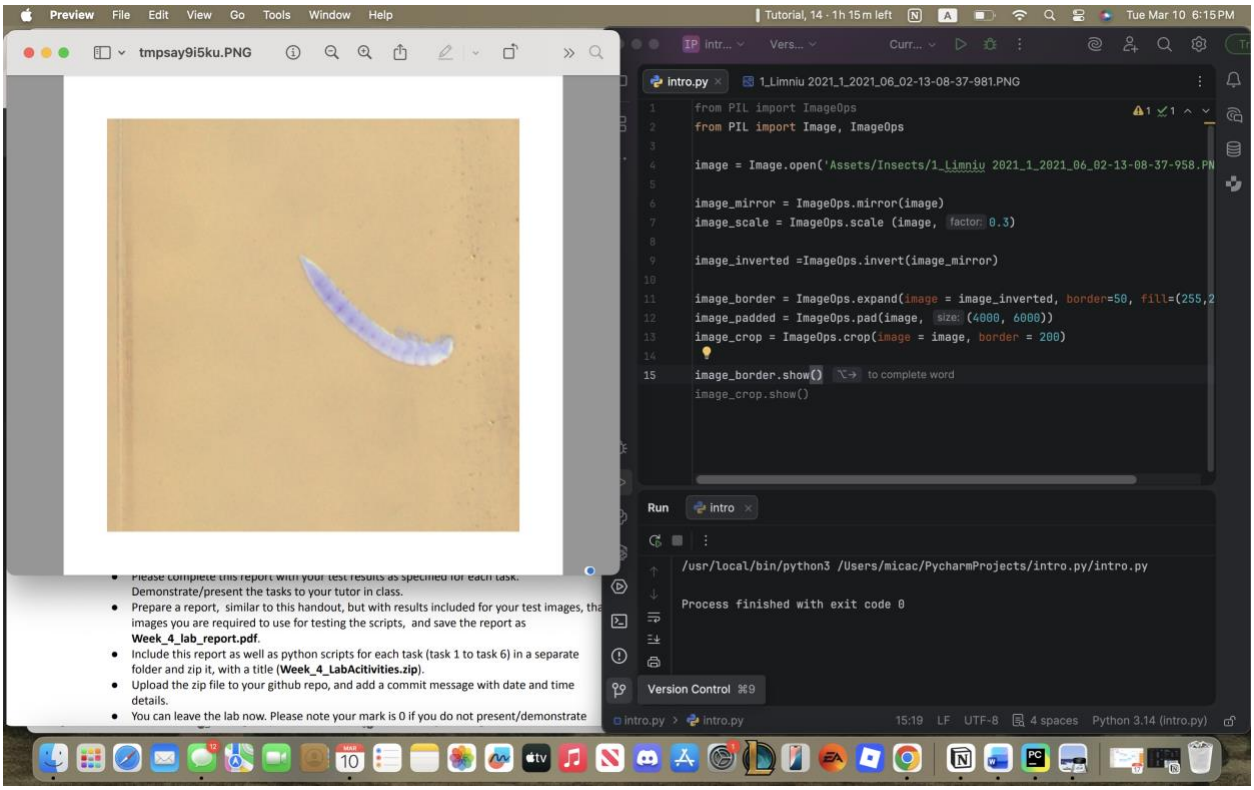
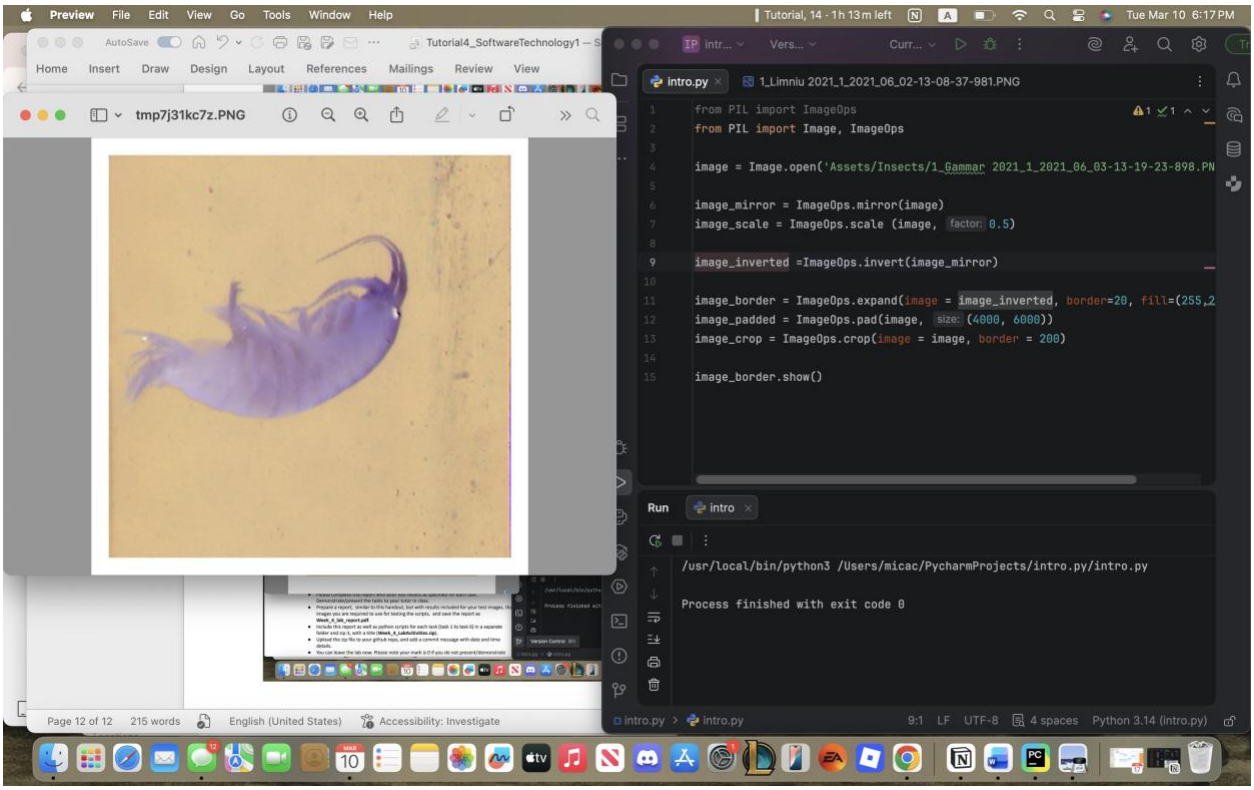
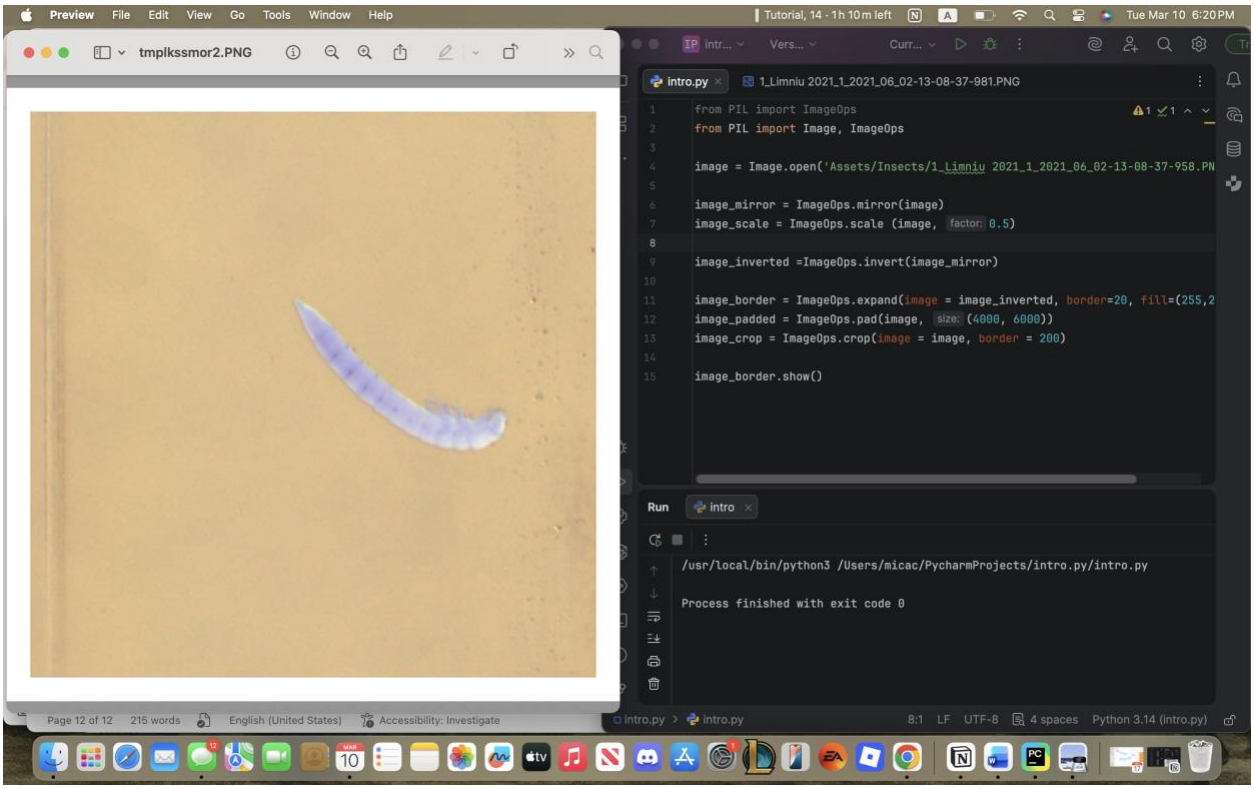
<u>Task 7:</u> Image Operations with Pillow/PIL library functions

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